# Data description to "CAMELS-DK: Hydrometeorological Time Series and Landscape Attributes for 3330 Catchments in Denmark"

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## Abstract

This document describes the CAMELS-DK dataset, which contains hydrometeorological time series and landscape attributes for 3,330 catchments in Denmark. The official data description paper has been submitted to Earth System Science Data (ESSD). This document provides a description of data processing methods and the structure of the dataset after downloading the unzipping.

## Background & Summary

CAMELS (Catchment Attributes and Meteorological Time Series for Large Samples) datasets have been developed in several countries and regions around the world, offering valuable data sources and testbeds for hydrological analysis and advancing data-driven hydrological modelling (Addor et al., 2017; Coxon et al., 2020; Höge et al., 2023). To address the lack of samples from lowland, groundwater-dominated, small-sized catchments, we have developed an extensive repository of a CAMELS-style dataset for Denmark (CAMELS-DK). As a particularity, CAMELS-DK not only contains data from 304 gauged basins, but also from a total of 3,300 largely ungauged catchments covering the entirety of Denmark. For all catchments, simulated variables from the National Hydrological Model of Denmark (DK-model) are included. Applications of CAMELS for hydrological modeling can be found in Liu et al. (2023) and Koch and Schneider (2022). We provide this dataset in an easy-to-use format, adhering to the Findable, Accessible, Interoperable, and Reusable (FAIR) principles.

## Methods

All data processing has been performed in python. The meteorological time series, DK-model simulations, and landscape attributes for each catchment are calculated using the python package *rioxarray*. The observed discharge and simulated discharge data are saved in .csv file, which is read by *pandas* package. The original raster dataset (*.tif*) is first read by *rioxarray* (https://corteva.github.io/rioxarray/stable/index.html#) and then clipped by the geometry of the catchment boundaries. The final values provided in CAMELS-DK represent the mean value of the clipped area, including pixels within and touching the catchment boundaries. This calculation routine is used for the attributes of topography, geology, and land use.

Climate data, DK-model simulations, and groundwater abstraction are stored in *.dfs* files, a format developed by DHI for the MIKE code. These files are first read by mikeio (https://github.com/DHI/mikeio) and converted to xarray. The catchment specific time series is then created with rioxarray and masked by the catchment boundaries on a daily basis and the spatial mean is calculated for each day. The static climate attributes are based on this time series, where the mean is calculated across all daily timesteps. For the attributes derived from DK-model simulations, we first calculate the statistics at each pixel across time and then compute the spatial average value for the corresponding catchments.

The hydrological signatures are calculated using the Toolbox for Streamflow Signatures in Hydrology (TOSSH) for both gauged and ungauged catchments based on the observed and simulated streamflow time series.

## Data Records

The data can be downloaded via https://doi.org/10.22008/FK2/AZXSYP. After downloading and unzipping the file, you will find four folders: *Shapefile*, *Attributes*, *Dynamics*, and *Python*.

- *Shapefile.* This folder contains the catchment boundaries ('CAMELS_DK_304_gauging_catchment_boundaries.shp') and the locations ('CAMELS_DK_304_gauging_stations.shp') of 304 gauged stations. It also includes the catchment boundaries for 3,330 ID15 catchments ('CAMELS_DK_3330_catchment_boundaries.shp'), allowing catchment boundaries to be delineated in Denmark. Please note that the boundaries of the 304 gauged catchments are delineated using the ID15 catchments shapefile, with the downstream catchment identified by its catch_id. The script for generating catchment boundaries is provided in the *Python* folder. Additionally, the folder contains the geological regions of Denmark, which are used in the discussion of spatial variability of landscape attributes in the ESSD paper.
- *Dynamics.* This folder contains two zipped files that need to be unzipped. Inside, you will find time series data for climate, DK-model simulated variables, discharge, and groundwater abstraction. The time series data for gauged and ungauged catchments are saved in separate folders. The data are stored in .CSV format for each catchment, with files identified by an 8-digit catch_id. Each file includes columns for time, catch_id, and the dynamic variables. The dynamic variables are listed in Appendix A. Via the catch_id, each time series can be linked to the shapefile.
- *Attributes.* This folder contains the attributes of 3,330 catchments, including both gauged and ungauged catchments. The data is saved in .CSV format. The landscape attributes include topography, climate indices, geology, land use derived from the basemap, land use based on CORINE, hydrological signatures based on observations, hydrological signatures based on simulations, and soil types derived from the Danish Soil Database and the European Soil Hydraulic Database. The first column of each table is catch_id, and the remaining columns contain the landscape attributes, which are summarized in Appendix B.
- *Python.* This folder provides code snippets for obtaining catchment boundaries based on the shapefile of 3,330 ID15 catchments. Additionally, it includes scripts for reading the .CSV files to extract landscape attributes and time series data for specific catchments.

The data are saved in the Dataverse repository, hosted by the Geological Survey of Denmark and Greenland (GEUS). The dataset can be accessed via this doi link: https://doi.org/10.22008/FK2/AZXSYP. The webpage provides data description, subject, keywords, related publications, and license information for the dataset. Additional details can be found by using the four button options: *Files*, *Metadata*, *Terms*, and *Versions*. The files can be viewed either in *Table* or *Tree* format under the *Files* section. The version history of the dataset is accessible under the *Versions* button.

## Version control

Dataverse automatically creates a new version of the dataset each time it is modified and republished. All versions of the dataset can be found by clicking the *Versions* button. While different versions of the dataset can be downloaded, the newest version is recommended. Below is a short summary of the different versions:

- *Version 1.0*: This is the initial version of the dataset, where the time series are saved in netCDF files.

- *Version 2.0*: Catchment boundaries and the locations of hydrological stations were added, and the time series were converted to .CSV format.
- *Version 3.0*: A dataset description is added. The column name of some static variables are changed.
- *Version 4.0*: The column name of some static variables is changed, the 'CAMELS_DK_signature_obs_based.csv' has been corrected so that it contains hydrological signatures based on the observed streamflow data at 304 hydrological stations. Catch_id = 426000042 has been modified, the value 0 during 01-01-2010 to 12-31-2011 has been removed.
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# Appendix

## Appendix A. Dynamic variables in CAMELS-DK

| Column name | Description | Unit | Sources |
|---|---|---|---|
| time | Time reference | d | |
| precipitation | Daily accumulated precipitation | $mm \cdot d^{-1}$ | Danish Meteorological Institute (Frie Data) |
| temperature | Daily mean temperature | °C | |
| pet | Daily potential evapotranspiration (Makkink) | $mm \cdot d^{-1}$ | |
| Qobs | Observed streamflow | $m^3 \cdot s^{-1}$ | overfladevandsdatabasen ODA |
| Qdkm | Simulated streamflow | $m^3 \cdot s^{-1}$ | DK-model (Stisen et al., 2020) |
| DKM_wcr | average water content in root zone | - | |
| DKM_dtp | phreatic depth to surface layer | m | |
| DKM_eta | actual evapotranspiration | $mm \cdot d^{-1}$ | |
| DKM_ rec | Total recharge to SZ | $mm \cdot d^{-1}$ | |
| DKM_sdr | SZ drainage flow from point | $m^3 \cdot s^{-1}$ | |
| DKM_sre | SZ exchange flow with river | $m^3 \cdot s^{-1}$ | |
| DKM_gwh | groundwater head in major deeper aquifer layers | m | |
| DKM_irr | Irrigation | $m^3 \cdot s^{-1}$ | |
| abstraction | Annual abstractions from groundwater | $m^3 \cdot s^{-1}$ | National well database |


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## Appendix B. Landscape attributes in CAMELS-DK.

Appendix B1. catchment location and topography

| Attribute name | Description | type/Unit | Data source |
|---|---|---|---|
| catch_id | Catchment identifier (eight-digit ID15 catchment name) | | Højberg et al.,(2021) |
| catch_outlet_lon | Catchment outlet longitude | | |
| catch_outlet_lat | Catchment outlet latitude | | |
| catch_flow_dir | Catchment flow direction | | |
| catch_accum_number | The total number of upstream accumulated ID15 catchment units | | |
| catch_area | Catchment accumulated area | $m^2$ | |
| gauged_type | Boolean values to indicate the catchment is gauged (True) or ungauged (False) | | |
| gauge_record_pct | If gauged catchment, the time percentage with available observed streamflow in the period from 1989-01-02 to 2019-12-31 | % | |
| elev_min | Minimum elevation of the catchment | m | The Danish Agency for Climate Data (Denmark's Height Model - Surface) |
| elev_median | Median elevation of the catchment | m | |
| elev_mean | Mean elevation of the catchment | m | |
| elev_max | Maximum elevation of the catchment | m | |
| slope_min | Minimum slope of the catchment | $m \cdot km^{-1}$ | |
| slope_median | Median slope of the catchment | $m \cdot km^{-1}$ | |
| slope_mean | Mean slope of the catchment | $m \cdot km^{-1}$ | |

| | | |
|---|---|---|
| slope_max | Maximum slope of the catchment | $m \cdot km^{-1}$ |
| pct_flat_area | Percentage of catchment area with slope smaller than 3m/km | % |

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169 Appendix B2. Climate indices in the dataset

| Attribute | Description | Unit |
|---|---|---|
| catch_id | Catchment identifier (eight-digit ID15 catchment name) | |
| p_mean | Mean of daily precipitation | $mm \cdot d^{-1}$ |
| t_mean | Mean of daily temperature | ◦C |
| pet_mean | Mean of daily potential evapotranspiration | $mm \cdot d^{-1}$ |
| aridity | aridity (PET /P, ratio of mean PET to mean precipitation) | - |
| p_seasonality | seasonality and timing of precipitation (estimated using sine curves to represent the annual temperature and precipitation cycles; positive (negative) values indicate that precipitation peaks in summer (winter); values close to 0 indicate uniform precipitation throughout the year) | - |
| frac_snow | fraction of precipitation falling as snow (i.e., on days colder than 0 ◦C) | % |
| high_prec_freq | frequency of high precipitation days (≥5 times mean daily precipitation) | $d \cdot yr^{-1}$ |
| high_prec_dur | average duration of high precipitation events (number of consecutive days ≥5 times mean daily precipitation) | d |
| high_prec_timing | season during which most high precipitation days (≥5 times mean daily precipitation) occur | season |
| low_prec_freq | frequency of dry days (< 1 mm/d) | $d \cdot yr^{-1}$ |
| low_prec_dur | average duration of dry periods (number of con- secutive days < 1 mm/d) | d |
| low_prec_timing | season during which most dry days (< 1 mm/d) occur | season |

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172 Appendix B3. Hydrographical signatures based on observed and simulated streamflow.

| Attribute | Description | Unit |
|---|---|---|
| catch_id | Catchment identifier (eight-digit ID15 catchment name) | |
| q_mean | mean daily streamflow | $mm \cdot d^{-1}$ |
| TotalRR | total runoff ratio (runoff divided by the precipitation) | |
| QP_elasticity | streamflow precipitation elasticity (sensitivity of streamflow to changes in precipitation at the annual timescale, using the mean daily streamflow as reference) | |
| FDC_slope | slope of the flow duration curve (between the log- transformed 33rd and 66th streamflow percentiles) | |
| BFI | baseflow index (ratio of mean daily baseflow to mean daily streamflow, hydrograph separation performed using the Ladson et al., 2013 digital filter | |
| HFD_mean | mean half-flow date (date on which the cumulative streamflow since October reaches half of the annual streamflow) | day of year |
| Q5 | 5% flow quantile (low flow | $mm \cdot d^{-1}$ |
| Q95 | 95% flow quantile (high flow | $mm \cdot d^{-1}$ |
| high_Q_frequency | frequency of high-flow days (> 9 times the median daily flow | |
| high_Q_duration | average duration of high-flow events (number of con- secutive days > 9 times the median daily flow | d |
| low_Q_frequency | frequency of low-flow days (< 0.2 times the mean daily flow | |
| low_Q_duration | average duration of low-flow events (number of consec- utive days < 0.2 times the mean daily flow | d |
| zero_Q_frequency | frequency of days with Q=0 | |
| SnowDayRatio | Snow day ratio | – |
| RLD | Rising limb density | $d^{-1}$ |
| AC1 | Lag-1 autocorrelation | – |
| RR_Seasonality | Runoff ratio seasonality | – |
| EventRR | Event runoff ratio | – |
| StorageFraction | Ratio between active and total storage | – |
| Recession_a_Seasonality | Seasonal variations in recession parameters | – |
| AverageStorage | Average storage from average baseflow and storage-discharge relationship | – |
| Spearmans_rho | Non-uniqueness in the storage-discharge relationship | – |
| EventRR_TotalRR_ratio | Ratio between event and total runoff ratio | – |
| VariabilityIndex | Variability index of flow | – |

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|---|---|---|
| IE_effect | Infiltration excess importance | – |
| IE_thresh | Infiltration excess threshold location (in a plot of quickflow volume vs. maximum intensity) | $mm \cdot d^{-1}$ |
| IE_thresh_signif | Infiltration excess threshold significance (in a plot of quickflow volume vs. maximum intensity) | – |
| SE_effect | Saturation excess importance | – |
| SE_thresh_signif | Saturation excess threshold significance (in a plot of quickflow volume vs. total precipitation) | – |
| SE_thresh | Saturation excess threshold location (in a plot of quickflow volume vs. total precipitation) | mm |
| SE_slope | Saturation excess threshold above-threshold slope (in a plot of quickflow volume vs. total precipitation) | – |
| Storage_thresh | Storage/saturation excess threshold location (in a plot of quickflow volume vs. antecedent precipitation index + total precipitation) | mm |
| Storage_thresh_signif | Storage/saturation excess threshold significance (in a plot of quickflow volume vs. antecedent precipitation index + total precipitation) | – |
| BaseflowMagnitude | Difference between maximum and minimum of annual baseflow regime | mm |
| ResponseTime | Catchment response time | d |
| FlashinessIndex | Richards-Baker flashiness idex | – |
| PQ_Curve | Slopes and breakpoints in cumulative P-Q regime curve | – |
| Q_n_day_max | n-day maximum streamflow | $mm \cdot d^{-1}$ |
| Q_skew | Skewness of streamflow | $mm^{3} \cdot day^{-3}$ |
| Q_var | Variance of streamflow | $mm^{2} \cdot day^{-2}$ |
| RecessionK_part | Recession constant of early/late (exponential) recessions | $1 \cdot d^{-1}$ |
| SeasonalTranslation | Amplitude ratio and phase shift between seasonal forcing and flow cycles | – |
| SnowStorage | Snow storage derived from cumulative P-Q regime curve | mm |
| zero_Q_duration | Zero flow duration | d |
| Q_7_day_min | 7-day minimum streamflow | $mm \cdot d^{-1}$ |
| CoV | Coefficient of variation | – |
| HFI_mean | Half flow interval | d |
| BaseflowRecessionK | Exponential recession constant | $1 \cdot d^{-1}$ |

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176 Appendix B4. Land use attributes.

| Attribute | Description | Years | Unit | Data source |
|---|---|---|---|---|
| catch_id | Catchment identifier (eight-digit ID15 catchment name) | | | |
| pct_ forest_corine_yyyy | Percentage of agricultural area | 1990, 2000, 2006, 2012, 2018 | % | Copernicus CORINE land cover (CLC, 2023) |
| pct_ crop_corine_yyyy | Percentage of grass and herb vegetation area | | % | |
| pct_ water_corine_yyyy | Percentage of medium-scale vegetation area | | % | |
| pct_ urban_corine_yyyy | Percentage of deciduous forest area | | % | |
| pct_ wetlands_corine_yyyy | Percentage of mixed forest area | | % | |
| pct_forest_levin_yyyy | Percentage of forest area | 2011, 2016, 2018, 2021 | % | Levin, G. 2022. Aarhus University, https://dce2.au.dk/pub/TR252.pdf |
| pct_ crop_levin_yyyy | Percentage of agricultural area | | % | |
| pct_ water_levin_yyyy | Percentage of lake and stream area | | % | |
| pct_ urban_levin_yyyy | Percentage of urban area | | % | |
| pct_ naturewet_levin_yyyy | Percentage of natural area (wet, agriculture, extensive) | | % | |
| pct_ naturedry_levin_yyyy | Percentage of natural area (dry, agriculture, extensive) | | % | |

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178 Appendix B5. Catchment attributes of soil types.
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| Attribute | Description | Unit | Data source |
|---|---|---|---|
| catch_id | Catchment identifier (eight-digit ID15 catchment name) | | |
| pct_sand | Sand content | % | Hiederer (2013a, b) |
| pct_silt | Silt content | % | |

| | | | |
|---|---|---|---|
| pct_clay | Clay content | % | |
| pct_organic | Organic carbon content | % | |
| bulk_density | Bulk density | $g \cdot cm^{-3}$ | |
| tawc | Total available water content (from PTR) | mm | |
| pct_gravel | Coarse Fragments | % | |
| root_depth | depth available for roots | m | |
| FC | water content at field capacity | $cm^3 \cdot cm^{-3}$ | Tóth et al. (2017) |
| HCC | hydraulic conductivity curve | $\log_{10}$ [$cm \cdot d^{-1}$ ] | |
| KS | saturated hydraulic conductivity | $\log_{10}[cm \cdot d^{-1}]$ | |
| MRC | moisture retention curve | $cm^3 \cdot cm^{-3}$ | |
| THS | Saturated water content | $cm^3 \cdot cm^{-3}$ | |
| WP | water content at wilting point | $cm^3 \cdot cm^{-3}$ | |
| pct_claynor_30 | Clay percentage in layer depth of 0-30cm | % | Adhikari et al.(2013) |
| pct_ claynor_60 | Clay percentage in layer depth of 30-60cm | % | |
| pct_ claynor_100 | Clay percentage in layer depth of 60-100cm | % | |
| pct_ claynor_200 | Clay percentage in layer depth of 100-200cm | % | |
| pct_ fsandno_30 | Fine sand percentage in layer depth of 0-30cm | % | |
| pct_ fsandno_60 | Fine sand percentage in layer depth of 30-60cm | % | |
| pct_ fsandno_100 | Fine sand percentage in layer depth of 60-100cm | % | |
| pct_ fsandno_200 | Fine sand percentage in layer depth of 100-200cm | % | |
| pct_ gsandno_30 | Coarse sand percentage in layer depth of 0-30cm | % | |
| pct_ gsandno_60 | Coarse sand percentage in layer depth of 30-60cm | % | |
| pct_ gsandno_100 | Coarse sand percentage in layer depth of 60-100cm | % | |
| pct_ gsandno_200 | Coarse sand percentage in layer depth of 100-200cm | % | |

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182 Appendix B6. Catchment attributes of geologic features

| Attribute | Description | Unit | Data source |
|---|---|---|---|
| catch_id | Catchment identifier (eight-digit ID15 catchment name) | | Stisen et al. (2020) and Ondracek (2023) |
| chalk_d | Depth to chalk | m | |
| uaquifer_t | Thickness of uppermost aquifer | m | |
| uaquifer_d | Depth to uppermost aquifer | m | |
| uclay_t | Thickness of uppermost clay | m | |
| usand_t | Thickness of uppermost sand | m | |
| pct_aeolain_sand | Percentage of aeolian sand, including dunes and cover sands | % | Pedersen et al. (2011a, b) |
| pct_water_deposit | Percentage of freshwater deposits | % | |
| pct_marsh | Percentage of marsh, alternating tin | % | |
| pct_marine_sand | Percentage of Marine sand and clay | % | |
| pct_beach | Percentage of Beach ridges | % | |
| pct_sandy_till | Percentage of Sandy and gravelly till | % | |
| pct_till | Percentage of till, clay and fine-sandy | % | |
| pct_glaf_sand | Percentage of glaciofluvial sand and gravel | % | |
| pct_glal_clay | Percentage of glaciolacustrine laminated clay, silt and fine sand | % | |
| pct_down_sand | Percentage of downwash sandy deposits | % | |

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|---|---|---|---|
| pct_glam_clay | Percentage of glaciomarine clay, silt and sand | % | |

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